INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE
(Affiliated to UEM Kolkata and Approved by AICTE)

Degree:**B.C.A**, Stream:**B.C.A**

End Semester Examination 2024 - Second Semester Examination,April-2024

Paper Code: BCACC201

Paper Name: Computer Architecture

Full Marks:100 Duration:3 Hours

Part - A
Attempt 10 questions
Each question carries 2 Marks (2 X 10)

1. Explain Instruction Formats. 2

or

1. Define Register Mode. 2

2. What are the major components of CPU? 2

or

2. Illustrate an example of a microoperation. 2

3. What are source program and object program in assembler? 2

or

3. What are high level programming languages? 2

4.	What is a buffer?	2
	or	
4.	Define assembly language program.	2
5.	Define Direct Memory Access.	2
	or	
5.	Explain Flynn's Classification.	2
6.	Define MIMD.	2
	or	
6.	What is the interrupt cycle?	2
7.	What is hit ratio?	2
	or	
7.	Define Vector Processor.	2
8.	What is tag field?	2
	or	
8.	Define Access time in terms of Auxiliary Memory.	2
9.	Define Implied Mode.	2

or

9. Describe how many multiplexers of what size are required to construct a common bus for 8 registers of 16 bits? 2

10. What is Control Word? 2

or

10. Find 10's complement of 3. 2

Part - B

Attempt 6 questions

Each question carries 5 Marks (5 X 6)

11. Assess 9's complement of (i) 00980100, (ii) 12349876 and (iii) 90009951 5

or

11. Assess 7's complement of (i) 5674, (ii) 3421 and (iii) 4022 5

12. What are the different components used in designing a basic computer of 4096 words? 5

or

12. Analyze $(-70)_{10} + (-80)_{10}$ with the help of 2's complement. 5

13. Demonstrate Internal interrupts. 5

or

13. Explain the memory read and memory write operation 5

14. What is the advantage of using Handshaking method over Strobe pulse method? 5

or

14. Describe Memory organization with the help of a block diagram. 5

15. Assess 16's complement of (i) 77C, (ii) 1B06 and (iii) B06. 5

or

15. Summarize arithmetic microoperations. 5

16. Illustrate control function with the help of a block diagram. 5

or

16. What is the direct and indirect addressing mode of instruction format? Illustrate with diagrams. 5

Part - C
Attempt 5 questions
Each question carries 10 Marks (10 X 5)

17. Illustrate three state buffer concepts with the Bus line. 10

or

17. Explain Zero-Address Instruction. Give example for explanation. 10

18. Draw and explain the flowchart for add and subtract operations. 10

or

18. Summarize the working of Booth Multiplication Algorithm with a flowchart and example. 10

19. Show the flowchart for the interrupt cycle and explain it. 10

or

19. Summarize the concept of 4 segments Pipelining with Space-Time Diagram. 10

20. Judge how the Auxiliary memory devices function. 10

or

20. Illustrate Associative memory working principle with an example. 10

21. Define Two-Address Instruction. Give example for explanation. 10

or

21. Evaluate X = (A + B) * (C + D) using one-address Instruction. 10